



EUNOIA JUNIOR COLLEGE
JC2 Preliminary Examinations 2021
General Certificate of Education Advanced Level
Higher 2

TITLE

CONFIDENTIAL INSTRUCTIONS

BIOLOGY

Paper 4 Practical

CONFIDENTIAL INSTRUCTIONS

9744/04

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2 hours 30 minutes

This document gives details of how to prepare for and administer the practical exam.

The information in this document and the identity of any materials are confidential and must NOT reach candidates either directly or indirectly.

This document consists of **4** printed pages.

Materials and apparatus for Question 1

Each candidate will need:

materials and apparatus for each candidate	quantity	✓
10% yeast suspension with 4% glucose in a beaker or container, labelled Y	at least 150 cm ³	
Hydrogencarbonate indicator solution in a beaker or container, labelled H , provided at room temperature	at least 40 cm ³	
1% ascorbic acid solution in a beaker or container, labelled C , provided at room temperature	at least 5 cm ³	
distilled water in a beaker or container, labelled W , provided at room temperature	at least 150 cm ³	
10 cm ³ syringes	3	
1 cm ³ syringe	1	
Test-tubes	6	
Boiling tube	1	
Bung with delivery tube, to fit boiling tube. Ensure air-tight seal around the junction of the delivery tube and the bung.	1	
Test-tube rack	1	
Beakers or containers, capacity 50–100 cm ³	5	
Beaker or container, capacity 400 cm ³ , labelled water-bath	1	
Beaker, capacity approximately 400 cm ³ , with approximately 200 cm ³ of water at 50–60 °C, and labelled hot water . The Supervisor may use a thermostatically controlled water-bath to provide this hot water for candidates.	1	
Beaker, capacity approximately 400 cm ³ , labelled tap water .	1	
Container with approximately 300 cm ³ tap water, labelled For washing	1	
Container, capacity approximately 500 cm ³ , labelled For waste	1	
Pasteur pipettes	3	
Glass rod	1	
White tile	1	
Paper towels	12	
Glass marker pen, permanent	1	
Stop-clock or timer showing seconds	1	
Thermometer, –10 °C to +110 °C	1	
Suitable eye protection	1	
5% yeast suspension with 5% glucose solution added in a beaker or container, labelled A1	at least 5 cm ³	
5% yeast suspension that has been boiled and cooled in a beaker or container, labelled A2	at least 5 cm ³	
5% yeast suspension made by mixing A1 and A2 in equal proportion, labelled A3	at least 5 cm ³	
1% methylene blue stain, freshly prepared, in a container with a pipette, labelled methylene blue	at least 5 cm ³	
Microscope with low-power (X10) and high-power (X40) objective lenses, eyepiece lens (X10), and eyepiece graticule fitted within the eyepiece	1	
Microscope slides and cover slips	3	

Preparation of materials

Test the yeast suspension well before the examination to make sure that the yeast cells are active. (Do not use yeast used for brewing as this does not always work actively enough in the time).

The day before the exam, the supervisor should check that the test-tubes do not have an acid residue that affects the colour of the indicator (see instructions for **H**).

- **Y**, 10% yeast suspension

10 g of dried yeast (for baking) is added to 80 cm³ of warm 4% glucose solution, stirred and made up to 100 cm³ with distilled water. 4% glucose solution can be made by adding 4 g of glucose to 100 cm³ of distilled water. As the yeast suspension may froth, it should be prepared in a large container.

When ready to put into beakers for each candidate, it is suggested that the yeast suspension is decanted into a new beaker or container, leaving the froth behind.

- **C**, 1% ascorbic acid solution

This is prepared by dissolving 1 g of ascorbic acid in 100 cm³ with distilled water.

- **H**, hydrogencarbonate indicator

This is prepared by putting 5 cm³ of the hydrogencarbonate indicator in a beaker or container and adding 95 cm³ distilled water. This should be kept covered. The indicator should be a red-brown colour at the start of the exam.

The supervisor should test a sample of the test-tubes to be used, to check the test-tubes do not have an acid residue that affects the indicator.

- Put a small volume of indicator into 3 test-tubes.
- The indicator should remain red-brown in the test-tubes
- If the indicator turns yellow, all of the test-tubes should be cleaned without leaving an acid residue.

The supervisor should test the indicator before the start of the exam by adding 1 drop of ascorbic acid solution into a test-tube with 5 cm³ of hydrogencarbonate indicator. The indicator should turn yellow.

- **A1**, 5% yeast suspension with 2% glucose solution

5g of dried yeast should be suspended in 100 cm³ of warm distilled water and then 2 g of glucose sprinkled over the surface and stirred.

This should be added at least 15 minutes before the candidate starts Question 1. The yeast needs to be actively frothing.

- **A2**, 5% yeast suspension that has been boiled and cooled

5g of dried yeast should be suspended in 100 cm³ distilled water and placed in a boiling water-bath for at least 5 minutes, then cooled to room temperature.

The Supervisor should check that all the yeast cells are dead by placing a drop of yeast suspension onto a microscope slide and add 1 drop of methylene blue as made up below. Mix with a glass rod and leave for 5 minutes. Add a coverslip and observe using high-power objective lens of the microscope. If the cells are not all blue boil for an extra five minutes.

- **A3**, 5% yeast suspension made by mixing A1 and A2 in equal proportions

After A2 has been made and **cooled to room temperature**, mix 50 cm³ of A1 with 50 cm³ of A2.

- 1% methylene blue solution

This is prepared by dissolving 1 g of methylene blue and 0.6 g of sodium chloride in 50 cm³ distilled water, stirred and made up to 100 cm³ with distilled water.

Materials and apparatus for Question 2

Each candidate will need:

materials and apparatus for each candidate	quantity	✓
Slide L1 (transverse section of <i>Cucurbita</i> stem)	1	
Microscope with low-power ($\times 10$) and high-power ($\times 40$) objective lenses, eyepiece lens ($\times 10$), and eyepiece graticule fitted within the eyepiece	1	

Microscopes should be provided at a ratio of one between two candidates.

To avoid confusion, only the lenses specified above are fitted in the microscopes to be used in the examination. Any lenses which are **not** $\times 10$ or $\times 40$ should be removed or replaced.

Organisation of the examination

All candidates must have access to the materials required for Question 1 throughout the whole period of the examination.

For Question 1(c) and Question 2, each candidate must have sole access to a microscope for an uninterrupted period of 1 hour 15 minutes only. Two candidates are not permitted to share the same microscope at the same time. Half of the candidates will have access to a microscope for a period of 1 hour 15 minutes from the start of the examination. These candidates should be advised to start with Question 1(c) and Question 2. Once candidates have finished working with the microscope, they can move on to the remaining parts of Question 1.

The other candidates should be advised to start with Question 1(a) and 1(b). After 1 hour 15 minutes, these candidates should be given access to the microscope and advised to move on to Question 1(c) and Question 2 as soon as possible.

Candidates should be advised that they can answer any part of the examination paper throughout the whole period of the examination. They will only have access to the microscope for Question 1(c) and Question 2 during the specified period of 1 hour 15 minutes.

Access arrangements to microscopes, including advice on which question to start with and timings, must be explained clearly to candidates before the start of the examination.